

Moein Razavi

AI/ML Engineer | Generative AI, Forecasting & Cloud Applications

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Professional Summary

AI/ML engineer and data scientist building AI applications and energy decision systems at Ford. Connects machine learning with deployed software: document RAG, text-to-SQL, utility-data ingestion, and forecasting for electricity-market bids. Brings Python/PyTorch expertise, cloud deployment experience, and full-stack engineering across energy and wearable health technology.

Professional Experience

Ford Motor Company

Data Scientist | Aug 2023 - Present

- **EV analytics assistant:** Designed an internal assistant for Ford EV teams to query analytics, legal documents, and vehicle manuals in natural language. Implemented specialist-agent routing with LangChain and GPT-4o/4o-mini, BM25-based retrieval-augmented generation (RAG), and schema-aware BigQuery SQL. Added conversational context, domain-rule answer review, and a Dash interface on Cloud Run with SAML/ADFS authentication.
- **Vehicle-to-grid market bidding:** Connected vehicle-availability forecasts to fleet energy-bid decisions using Python/PyTorch. Selected PatchTST through model comparisons and tuned candidates with Optuna on Vertex AI GPUs. Combined ERCOT prices via gridstatus, Monte Carlo capacity estimates, and greedy vehicle selection; implemented load reallocation when vehicles disconnect.
- **Utility data onboarding (HOMeload):** Designed and deployed a Python/FastAPI pipeline that converts heterogeneous customer electricity records into standardized inputs for energy analysis. Combined Gemini schema discovery with pandas/lxml extraction, deterministic validation, up to 3 corrective attempts, and BigQuery MERGE for repeatable loads; deployed with Docker, Cloud Run, and Terraform.
- **Rate recommendation:** Built an application to compare utility plans using XGBoost electricity-bill predictions from tariffs, vehicle telemetry, and household consumption. Delivered Flask APIs, BigQuery workflows, and a SAML-authenticated Dash interface; automated Cloud Run deployments through Tekton CI/CD.
- **Energy data engineering and reporting:** Automated preparation of energy-simulator inputs and processing of optimizer outputs with PySpark, BigQuery, and Airflow. Exposed discharge results through Looker Studio reporting and REST APIs so teams could review simulated and operational energy performance.
- **Code analysis and cloud cost estimation:** Built a FastAPI/LangChain tool for engineers to inspect code complexity and estimate Vertex AI compute costs. Used GitPython, AST parsing, and streamed results via Server-Sent Events; calculated cost estimates from LLM-generated resource assumptions and static pricing.

CereVu Medical, Inc.

Software Developer / Research Engineer (Concurrent Contract) | Jun 2023 - Sep 2024

- **Wearable health platform:** Developed Vitality and Clinical, two SwiftUI apps for capturing and reviewing 9 physiological signals through CoreBluetooth BLE and Swift Charts. Migrated Clinical from UIKit to MVVM-Coodinator; implemented calibration, exports, role-based workflows, and OAuth 2.0/JWT authentication.
- **Backend and delivery:** Connected wearable data to a React/Next.js dashboard through Python/Flask services, WebSocket streaming, and MongoDB. Deployed Docker services on AWS EC2 with GitHub Actions and distributed iOS builds through TestFlight.

Education

Ph.D., Industrial Engineering | Texas A&M University | Dec 2023

Master of Engineering, Computer Engineering | Texas A&M University | May 2021

Selected Projects

- **Business Action Prioritizer:** Designed a Google ADK/Gemini workflow that turns business data and scoring rubrics into ranked actions with supporting analysis. Gated PandasAI execution on user approval; implemented FastAPI/SQLite sessions and MCP report delivery through Cloud Storage.
- **InspecaLytics:** Built an inspection workflow from annotated photo capture to Word/PowerPoint reports using SwiftUI, React/TypeScript, and Supabase. Implemented field-level conflict resolution for offline edits, ARKit/Apple Vision photo organization, and XCTest/Vitest tests for application components.
- **Clink Social:** Built location-based discovery and messaging with SwiftUI, MapKit, and Firebase Auth/Firestore/FCM. Implemented server-side feed aggregation and notifications in TypeScript Cloud Functions, plus QR-based cross-device sign-in with time-limited OAuth/JWT tokens.

Research Experience

Texas A&M University

Research Assistant | Sep 2018 - May 2024

Published research in machine learning and sensing; 500+ Google Scholar citations (September 2026).

- **Wearable stress detection:** Compared classifiers including XGBoost on windowed heart-rate/motion data from 54 students over 40 days, labeled by self-reports. Used Python, cross-validation, ROC-AUC evaluation, and SHAP interpretation.
- **MERCURY:** Co-authored research using random projection with quantization and cached computation reuse; reported 1.97x average training speedup across 12 models with accuracy similar to baseline.
- **OpenSync:** Co-developed open-source Python software using Lab Streaming Layer to synchronize sensors and experiment events; published tests measured RMS recording-time lag below 0.07 ms. Code
- **Computer vision:** Developed face-mask and physical-distance monitoring using TensorFlow Faster R-CNN detectors and calibrated perspective transforms to measure distances in video.
- **GlucoseCoach:** Built an iOS app and Python/Flask backend for personalized glucose monitoring.

Technical Skills

Core Languages: Python, SQL, TypeScript/JavaScript, Swift; additional: C++, Java, R, MATLAB

ML Frameworks: PyTorch, TensorFlow, scikit-learn, XGBoost, Hugging Face, Optuna, OpenCV

ML Methods: Time-series forecasting, PatchTST, Monte Carlo simulation, cross-validation, SHAP/LIME

Generative AI: LangChain, Google ADK, Gemini, RAG, BM25, agent routing, text-to-SQL, MCP, PandasAI

Data: BigQuery, PySpark, Airflow, pandas, NumPy, PostgreSQL, MongoDB, Firebase, Supabase

Software Engineering: FastAPI, Flask, REST, WebSocket, React/Next.js, SwiftUI, CoreBluetooth, OAuth/JWT/SAML

Cloud: GCP (Vertex AI, Cloud Run, Dataproc), AWS (EC2, S3, Lambda, SageMaker), Azure OpenAI

Delivery and Testing: Docker, Terraform, Kubernetes, Git, GitHub Actions, Tekton, AWS CodeDeploy, Fastlane, TestFlight, XCTest, Vitest

Selected Publications

Razavi, M., et al. Machine Learning, Deep Learning, and Data Preprocessing Techniques for Detecting, Predicting, and Monitoring Stress and Stress-Related Mental Disorders: Scoping Review. JMIR Mental Health, 2024.

Janfaza, V., Weston, K., Razavi, M., et al. MERCURY: Accelerating DNN Training By Exploiting Input Similarity. IEEE HPCA, 2023.

Razavi, M., et al. OpenSync: An open-source platform for synchronizing multiple measures in neuroscience experiments. Journal of Neuroscience Methods, 2022.

Razavi, M., et al. An Automatic System to Monitor the Physical Distance and Face Mask Wearing of Construction Workers in COVID-19 Pandemic. SN Computer Science, 2022.